

The 6 Parent Functions

Name _____ · Aug 10

Formula bank **Linear:** $f(x) = x$ **Quadratic:** $f(x) = x^2$ **Cubic:** $f(x) = x^3$
Absolute value: $f(x) = |x|$ **Square root:** $f(x) = \sqrt{x}$ **Reciprocal:** $f(x) = \frac{1}{x}$

VOCABULARY

Parent function — the _____ member of a family of functions --- no shifts, stretches, or reflections applied.

Domain — the set of all allowed _____ (x) values a function can use.

Range — the set of all _____ (y) values a function actually produces.

EXAMPLE 1 · TYPE 1 — IDENTIFY FAMILY, DOMAIN, RANGE

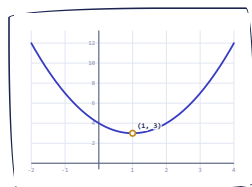
Identify the parent function of $f(x) = \frac{1}{x}$ and give its domain and range.

1. The rule is a fraction with x in the denominator, so this is the _____ parent function.
2. A denominator can never equal 0, so x cannot be 0: domain is _____.
3. Since $y = 1/x$ never produces an output of exactly 0 either, the range is _____.

EXAMPLE 2 · TYPE 2 — SHIFTED EQUATION, NAME THE PARENT

Name the **ORIGINAL** parent family of $g(x) = (x - 1)^2 + 3$.

1. The -1 and $+3$ only _____ — they never change the family.
2. The core operation is _____, so the parent is the _____ family: $f(x) = x^2$.



YOU TRY

1.

Identify the parent function of $f(x) = x^3$ and give its domain and range.

2.

Identify the parent function of $f(x) = |x|$ and give its range.

SUMMARY

To identify a parent function: match the _____ to one of the 6 known shapes, then check for _____ caused by a root or a denominator.

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EXAMPLE 1 · TYPE 1 — IDENTIFY FAMILY, DOMAIN, RANGE

Identify the parent function of $f(x) = \frac{1}{x}$ and give its domain and range.

1. The rule is a fraction with x in the denominator, so this is the _____ parent function.
2. A denominator can never equal 0, so x cannot be 0: domain is _____.
3. Since $y = 1/x$ never produces an output of exactly 0 either, the range is _____.

EXAMPLE 2 · TYPE 2 — SHIFTED EQUATION, NAME THE PARENT

Name the **ORIGINAL** parent family of $g(x) = (x - 1)^2 + 3$.

1. The -1 and $+3$ only _____ — they never change the family.
2. The core operation is _____, so the parent is the _____ family: $f(x) = x^2$.



YOU TRY

1.

Identify the parent function of $f(x) = x^3$ and give its domain and range.

2.

Identify the parent function of $f(x) = |x|$ and give its range.

SUMMARY

To identify a parent function: match the _____ to one of the 6 known shapes, then check for _____ caused by a root or a denominator.

Meet the Parent Functions

Name: _____

Period: ____

Date: Aug 10

FINAL ANSWERS — RECORD HERE

1. _____ 2. _____ 3. _____ 4. _____ 5. _____ 6. _____ 7. _____
8. _____ 9. _____ 10. _____ 11. _____

For each equation, name the parent function family, then state its domain and range using inequality notation.

TYPE 1 Identify the family, domain, and range

EXAMPLE

Example: $f(x) = \frac{1}{x}$ is the **reciprocal** parent. A denominator can't be 0, so the domain is $x \neq 0$; the graph never touches $y = 0$ either, so the range is $y \neq 0$.

Directions: Directions: Name the parent family, then give the domain and range.

1. $f(x) = x$

2. $f(x) = x^2$

3. $f(x) = x^3$

4. $f(x) = |x|$

5. $f(x) = \sqrt{x}$

6. $f(x) = \frac{1}{x}$

TYPE 2 Match a shifted equation back to its parent family

EXAMPLE

Example: $g(x) = \sqrt{x} + 4$ still has a radical, so its parent family is **square root** (the +4 just shifts it up 4).

Directions: Even though these are shifted, name the **ORIGINAL** parent family only.

7. $g(x) = (x - 2)^2$

8. $g(x) = |x + 1| - 3$

9. $g(x) = \frac{1}{x - 5}$

10. $g(x) = (x + 3)^3 + 1$
